

Business Problem Statement

Pharmaceutical Supply Chain Optimization

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Domain: Healthcare / Pharmaceutical Supply Chain Analytics

Tools: Python, SQL (MySQL), Power BI, Excel

1. Background

Pharmaceutical companies and distributors must continuously balance two competing risks: running out of critical medicines (stockouts) and holding excess inventory that ties up capital and risks expiry (overstocking). Both failure modes carry serious consequences in this industry. A stockout of Insulin or Metformin can directly endanger patient health, while chronic overstocking increases holding costs and waste. This project analyzes a simulated pharmaceutical supply chain dataset covering four widely prescribed drugs (Metformin, Lisinopril, Insulin, and Atorvastatin) to quantify how well forecasted demand is currently being matched to planned stock levels.

2. Problem Statement

The organization does not have a clear, data-driven view of how its inventory planning is performing against demand. Across 100,000 Stock Keeping Unit-level demand and stocking records, it is unknown:

- How often planned stock levels fail to cover forecasted demand (understock risk), and how often they exceed it (overstock waste).
- Whether the current restocking cadence — Weekly, Monthly, or Quarterly — has any measurable effect on reducing stockout risk.
- Which drugs are most exposed to shortage risk and would benefit most from tighter forecasting or safety-stock policy.
- How large the financial and operational exposure is, in absolute units, from both understocking and overstocking.

Without answering these questions, planners are making restocking decisions on intuition rather than evidence, which risks both patient-safety incidents (from stockouts) and unnecessary carrying costs (from overstocking).

3. Business Objectives

- Quantify the current scale of understock vs. overstock across the drug portfolio.
- Determine whether restocking strategy (Weekly / Monthly / Quarterly) measurably changes stockout risk.
- Identify the drugs and stock bands most at risk of shortage so they can be prioritized for tighter monitoring.
- Build a reusable SQL + Python analysis pipeline and an interactive Power BI dashboard that stakeholders can use to monitor stock health going forward.

4. Scope

The analysis covers demand forecasts and optimal stock levels for 4 drugs across 100,000 records, each tagged with a restocking strategy. It includes data cleaning, derived-metric engineering (Stock_Difference, Stock_Status, Priority), exploratory data analysis in Python, relational querying in MySQL, and dashboarding in Power BI. It does not cover real-time inventory tracking, supplier lead times, or cost/pricing data, as these were not present in the source dataset.

5. Success Criteria

- A clear, quantified breakdown of understock vs. overstock across the dataset.
- A comparison of stockout risk across restocking strategies, with a data-backed conclusion on whether cadence matters.
- A ranked view of the most critical (highest-shortage) drugs and Stock Keeping Units.
- An interactive Power BI dashboard and a documented SQL/Python analysis that stakeholders can reuse.

6. Stakeholders

- Supply Chain / Inventory Planning Team — primary users of the recommendations.
- Pharmacy / Clinical Operations — impacted by stockout risk for critical drugs.
- Finance — impacted by overstock carrying cost.
- Data/Analytics Team — owns the pipeline and dashboard going forward.